

Results 1 - A Rectangular Tank

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1 Results and Discussion

In this section, we will demonstrate the accuracy of our method in geometries where there exists an analytical solution for the sloshing modes and frequencies. In particular, we will calculate the absolute error between the numerical and analytical eigenvalues (i.e $|\lambda_a - \lambda_n|$, where λ_a and λ_n are the analytical and numerical eigenvalues respectively).

1.1 Normalising the Numerical Eigenfunctions

Notice that as a result of the linearity of Laplace's equation and the boundary conditions, ϕ is define up to a multiplicative constant. As a result, in order to compare our eigenfunctions that we determined numerically to the analytical eigenfunctions, we normalise the numerical eigenfunctions, so that they are order 1. We first take 100 points inside the triangle (that are a distance of 10^{-4} away from the corners) and then calculate the average value of the velocity potential at these points and denote it as: $\langle \hat{\Phi} \rangle$. Then we can calculate the normalised eigenfunctions, denoted as Φ , as follows:

$$\Phi = \frac{\hat{\Phi}}{\langle \hat{\Phi} \rangle}, \quad (1)$$

where $\hat{\Phi}$ denotes the unnormalised eigenfunctions.

1.2 A Rectangular Tank

The sloshing modes and frequencies of a $l \times h$ rectangle can be determined analytically using the technique of separation of variables. The details of the full calculation can be found in [1]. Here we consider our rectangle to be centred at $(x, y) = (0, -h/2)$ with the undisturbed free surface along the x-axis. The

sloshing modes and their corresponding eigenvalues are given below:

$$\phi_k = \cos\left(\frac{\pi k}{2l}(x+l)\right) \cosh\left(\frac{\pi k}{2l}(y+h)\right) \quad (2)$$

$$\lambda_k = \frac{\pi k}{2l} \tanh\left(\frac{\pi k h}{2l}\right). \quad (3)$$

Table 1 compares the absolute error between the analytical and numerical eigenvalues. Here the size of the matrix A is 1516×101 . In addition, the numerical eigenvalues in table 1 were obtained by excluding the Newman part in the basis functions, as it was found to increase the accuracy.

Table 1: In this table we compare the errors between the analytical and numerical sloshing frequencies for rectangular tanks with different length to breadth ratios.

Sloshing Mode	Absolute Error		
	$h/l=0.5$	$h/l=1$	$h/l=2$
0	10^{-15}	10^{-15}	10^{-15}
1	10^{-12}	10^{-14}	10^{-13}
2	10^{-15}	10^{-14}	10^{-14}
3	10^{-13}	10^{-12}	10^{-11}
4	10^{-12}	10^{-14}	10^{-11}
5	10^{-13}	10^{-12}	10^{-9}
6	10^{-11}	10^{-13}	10^{-9}
7	10^{-9}	10^{-12}	10^{-6}
8	10^{-9}	10^{-10}	10^{-6}
9	10^{-8}	10^{-8}	10^{-4}
10	10^{-7}	10^{-8}	10^{-4}

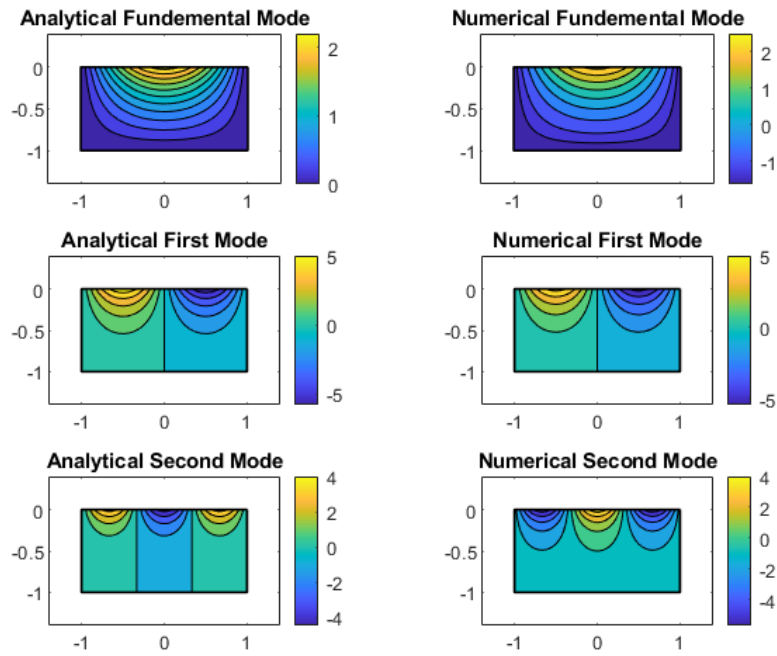


Figure 1: This figure compares the contour plots of the analytical and numerical streamfunction. It can be seen that the modes do not reflect the accuracy of the eigenvalues, especially for higher modes.

References

- [1] R.A. Ibrahim. *Liquid Sloshing Dynamics: Theory and Applications*. Cambridge University Press, 2005. ISBN: 9781139444132.